EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 6.01

C Sine & Cosine Rules, 3D Trigonometry & Vector Proofs

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

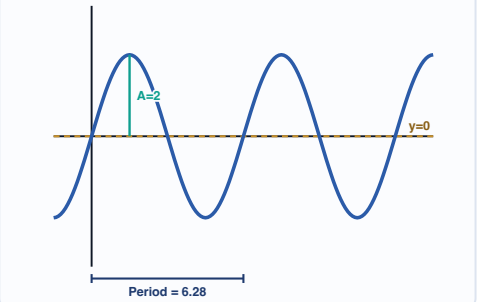
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 B	Q2 C	Q3 D	Q4 A	Q5 B
Q6 C	Q7 D	Q8 A	Q9 A	Q10 B
Q11 A	Q12 C	Q13 C	Q14 D	Q15 B
Q16 D	Q17 D	Q18 A	Q19 A	Q20 C
Q21 D	Q22 C	Q23 B	Q24 D	Q25 C

Sine & Cosine Rules, 3D Trigonometry & Vector Proofs Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 16x + 80$ in the domain of Sine/Cosine Rules, 3D Trigonometry & Vectors. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(10)$.



A 44

B 36 ✓

C 20

D 34

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Complete the square: $x^2 - 16x + 80 = (x-8)^2 - 64 + 80 = (x-8)^2 + 16$, giving minimum value $k = 16$. **[M1]**
2. Evaluate at $x = 10$: $f_1(10) = (2)^2 + 16 = 20$. **[M1]**
3. Sum the two quantities: $k + f_1(10) = 16 + 20 = 36$. **[A1]**

- **Final Verified Answer:** 36.

2 MCQ

A mathematical model in Sine/Cosine Rules, 3D Trigonometry & Vectors is defined by the cubic function $y_2(x) = 4x^3 - 8x^2 + 15x - 13$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123

B 60

C 75 ✓

D 35

WORKED SOLUTION & VERIFICATION

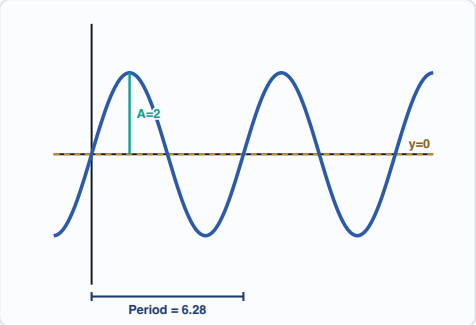
Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 16x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 12(9) - 16(3) + 15 = 75$. **[M1][A1]**

- **Final Verified Answer:** 75.

3 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, two displacement vectors are given by $\mathbf{u}_3 = \langle 8, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 16

B 128

C 256

D 144 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (8)(9) + (-8)(7) = 72 + (-56) = 16$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 8^2 + (-8)^2 = 128$. **[M1]**
3. Combine: $16 + 128 = 144$. **[A1]**

- **Final Verified Answer:** 144.

4 MCQ

The linear growth profile shown in the diagram for Sine/Cosine Rules, 3D Trigonometry & Vectors follows an arithmetic progression with first term $a_1 = 9$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

- **A** 765 ✓
- **B** 93
- **C** 855
- **D** 1395

WORKED SOLUTION & VERIFICATION

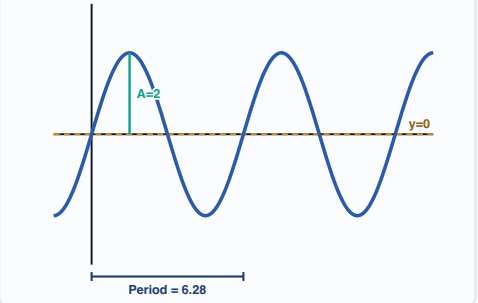
Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 9, d = 6$: $S_{15} = \frac{15}{2}[18 + 14 \times 6] = \frac{15}{2} \times 102 = 765$. **[M1][A1]**

- **Final Verified Answer:** 765.

5 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 32x)\,\mathrm{d}x$.



- **A** 136
- **B** 112 ✓
- **C** 144
- **D** 57

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Integrate term by term: $\int (18x^2 + 32x)\,\mathrm{d}x = [6x^3 + 16x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 16(2)^2) - 0 = 48 + 64 = 112$. **[M1][A1]**

- **Final Verified Answer:** 112.

6 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, consider the quadratic function $f_6(x) = x^2 - 18x + 169$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(16)$.

A 234

B 137

C 225 ✓

D 218

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Complete the square: $x^2 - 18x + 169 = (x-9)^2 - 81 + 169 = (x-9)^2 + 88$, giving minimum value $k = 88$. **[M1]**
2. Evaluate at $x = 16$: $f_6(16) = (7)^2 + 88 = 137$. **[M1]**
3. Sum the two quantities: $k + f_6(16) = 88 + 137 = 225$. **[A1]**

- **Final Verified Answer:** 225.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 9x^2 + 33x - 20$ studied in Sine/Cosine Rules, 3D Trigonometry & Vectors. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 801

B 624

C 111

D 657 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 18x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 12(64) - 18(8) + 33 = 657$. **[M1][A1]**

- **Final Verified Answer:** 657.

8 MCQ

In Sine/Cosine Rules, 3D Trigonometry & Vectors, two displacement vectors are given by $\mathbf{u}_8 = \langle 9, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 167 ✓

B 37

C 130

D 237

WORKED SOLUTION & VERIFICATION

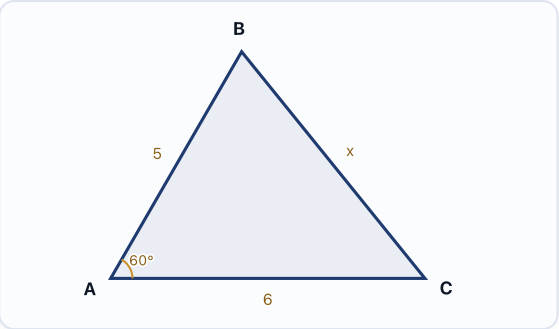
Mark Scheme & Examiner Solution (Sine/Cosine Rules, 3D Trigonometry & Vectors):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (9)(8) + (-7)(5) = 72 + (-35) = 37$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 9^2 + (-7)^2 = 130$. **[M1]**
3. Combine: $37 + 130 = 167$. **[A1]**

- **Final Verified Answer:** 167.

9 MCQ

In $\triangle ABC$ (#1), $AB = 5$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



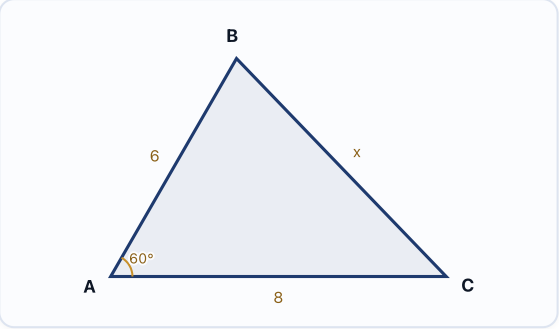
- **A** $\sqrt{31}$ cm ✓
- **B** 8 cm
- **C** $\sqrt{91}$ cm
- **D** $\sqrt{61}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 5^2 + 6^2 - 2(5)(6)\cos 60^\circ = 25 + 36 - 30 = 31$, so $BC = \sqrt{31}$ cm.

10 MCQ

In $\triangle ABC$ (#2), $AB = 6$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



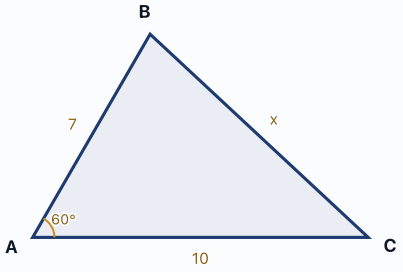
- **A** $\sqrt{148}$ cm
- **B** $\sqrt{52}$ cm ✓
- **C** $\sqrt{100}$ cm
- **D** 11 cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 6^2 + 8^2 - 2(6)(8)\cos 60^\circ = 36 + 64 - 48 = 52$, so $BC = \sqrt{52}$ cm.

11 MCQ

In $\triangle ABC$ (#3), $AB = 7$ cm, $AC = 10$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



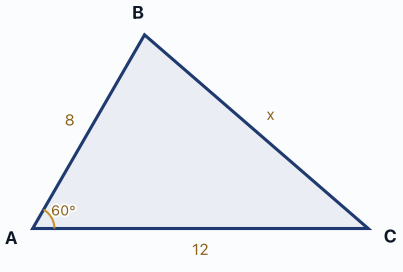
- **A** $\sqrt{79}$ cm ✓
- **B** $\sqrt{149}$ cm
- **C** $\sqrt{219}$ cm
- **D** 14 cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 7^2 + 10^2 - 2(7)(10)\cos 60^\circ = 49 + 100 - 70 = 79$, so $BC = \sqrt{79}$ cm.

12 MCQ

In $\triangle ABC$ (#4), $AB = 8$ cm, $AC = 12$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



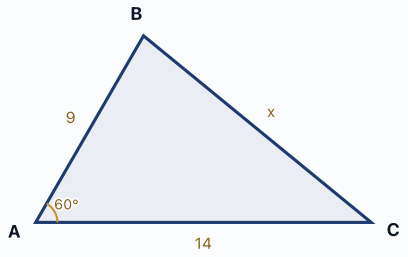
- **A** $\sqrt{304}$ cm
- **B** 17 cm
- **C** $\sqrt{112}$ cm ✓
- **D** $\sqrt{208}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 8^2 + 12^2 - 2(8)(12)\cos 60^\circ = 64 + 144 - 96 = 112$, so $BC = \sqrt{112}$ cm.

13 MCQ

In $\triangle ABC$ (#5), $AB = 9$ cm, $AC = 14$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{403}$ cm

B $\sqrt{277}$ cm

C $\sqrt{151}$ cm ✓

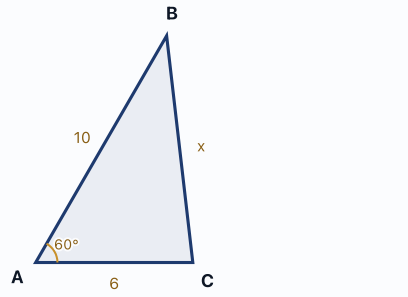
D 20 cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 9^2 + 14^2 - 2(9)(14)\cos 60^\circ = 81 + 196 - 126 = 151$, so $BC = \sqrt{151}$ cm.

14 MCQ

In $\triangle ABC$ (#6), $AB = 10$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 13 cm

B $\sqrt{196}$ cm

C $\sqrt{136}$ cm

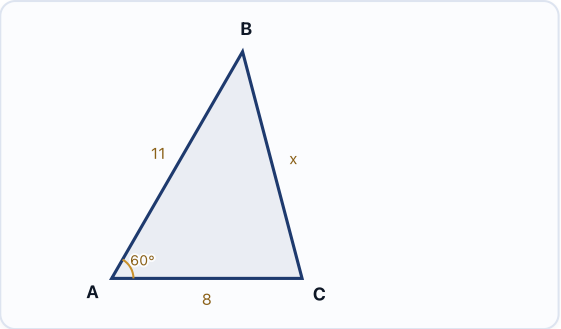
D $\sqrt{76}$ cm ✓

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 10^2 + 6^2 - 2(10)(6)\cos 60^\circ = 100 + 36 - 60 = 76$, so $BC = \sqrt{76}$ cm.

15 MCQ

In $\triangle ABC$ (#7), $AB = 11$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{273}$ cm

B $\sqrt{97}$ cm ✓

C 16 cm

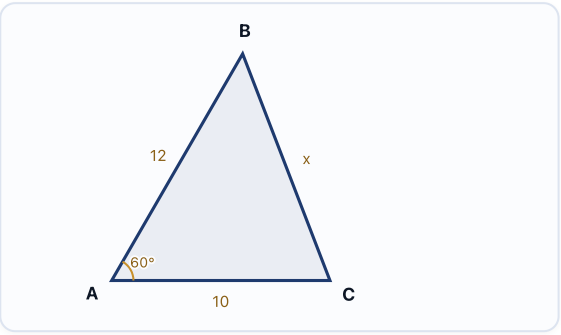
D $\sqrt{185}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 11^2 + 8^2 - 2(11)(8)\cos 60^\circ = 121 + 64 - 88 = 97$, so $BC = \sqrt{97}$ cm.

16 MCQ

In $\triangle ABC$ (#8), $AB = 12$ cm, $AC = 10$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 19 cm

B $\sqrt{364}$ cm

C $\sqrt{244}$ cm

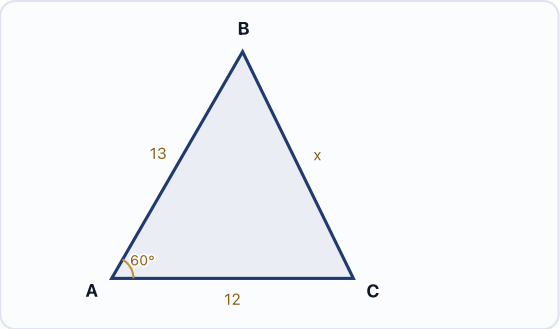
D $\sqrt{124}$ cm ✓

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 12^2 + 10^2 - 2(12)(10)\cos 60^\circ = 144 + 100 - 120 = 124$, so $BC = \sqrt{124}$ cm.

17 MCQ

In $\triangle ABC$ (#9), $AB = 13$ cm, $AC = 12$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{469}$ cm

B 22 cm

C $\sqrt{313}$ cm

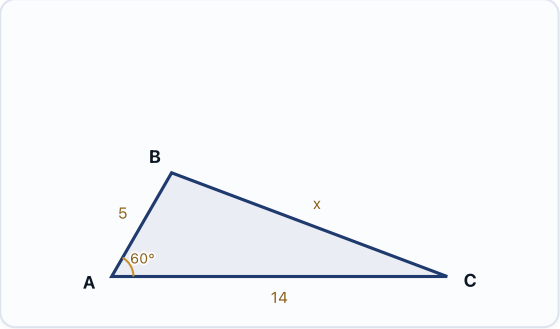
D $\sqrt{157}$ cm ✓

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 13^2 + 12^2 - 2(13)(12)\cos 60^\circ = 169 + 144 - 156 = 157$, so $BC = \sqrt{157}$ cm.

18 MCQ

In $\triangle ABC$ (#10), $AB = 5$ cm, $AC = 14$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{151}$ cm ✓

B $\sqrt{221}$ cm

C 16 cm

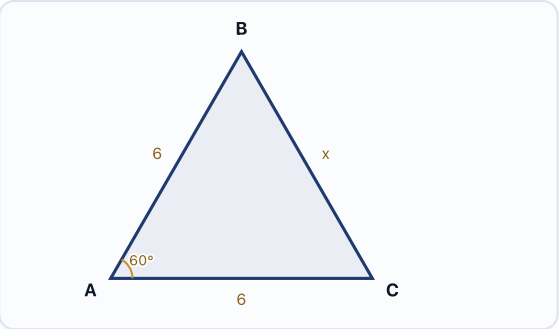
D $\sqrt{291}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 5^2 + 14^2 - 2(5)(14)\cos 60^\circ = 25 + 196 - 70 = 151$, so $BC = \sqrt{151}$ cm.

19 MCQ

In $\triangle ABC$ (#11), $AB = 6$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



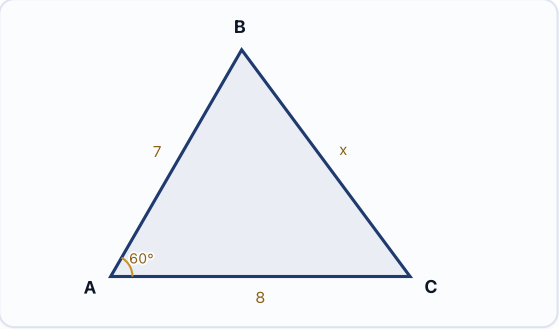
- **A** $\sqrt{36}$ cm ✓
- **B** $\sqrt{72}$ cm
- **C** 9 cm
- **D** $\sqrt{108}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 6^2 + 6^2 - 2(6)(6)\cos 60^\circ = 36 + 36 - 36 = 36$, so $BC = \sqrt{36}$ cm.

20 MCQ

In $\triangle ABC$ (#12), $AB = 7$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



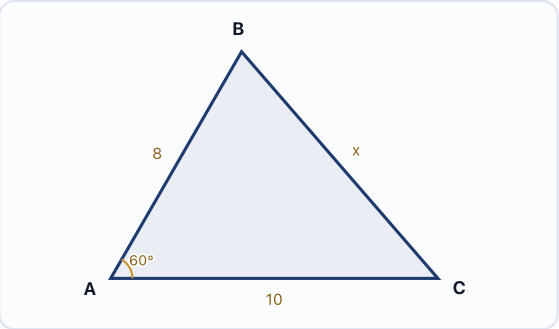
- **A** 12 cm
- **B** $\sqrt{169}$ cm
- **C** $\sqrt{57}$ cm ✓
- **D** $\sqrt{113}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 7^2 + 8^2 - 2(7)(8)\cos 60^\circ = 49 + 64 - 56 = 57$, so $BC = \sqrt{57}$ cm.

21 MCQ

In $\triangle ABC$ (#13), $AB = 8$ cm, $AC = 10$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{244}$ cm

B $\sqrt{164}$ cm

C 15 cm

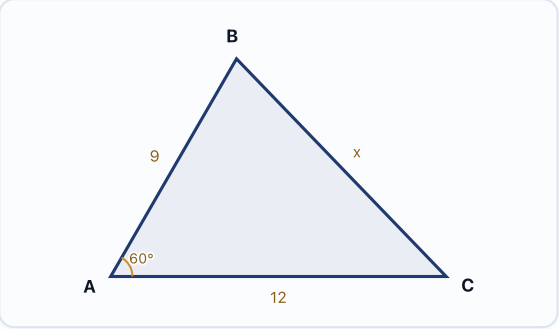
D $\sqrt{84}$ cm ✓

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 64 + 100 - 80 = 84$, so $BC = \sqrt{84}$ cm.

22 MCQ

In $\triangle ABC$ (#14), $AB = 9$ cm, $AC = 12$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{333}$ cm

B $\sqrt{225}$ cm

C $\sqrt{117}$ cm ✓

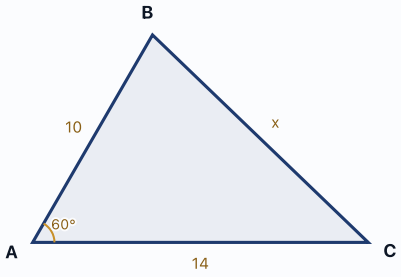
D 18 cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 9^2 + 12^2 - 2(9)(12)\cos 60^\circ = 81 + 144 - 108 = 117$, so $BC = \sqrt{117}$ cm.

23 MCQ

In $\triangle ABC$ (#15), $AB = 10$ cm, $AC = 14$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A $\sqrt{296}$ cm

B $\sqrt{156}$ cm ✓

C 21 cm

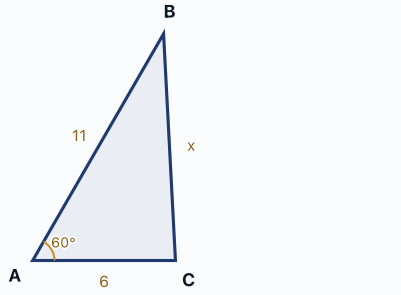
D $\sqrt{436}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 10^2 + 14^2 - 2(10)(14)\cos 60^\circ = 100 + 196 - 140 = 156$, so $BC = \sqrt{156}$ cm.

24 MCQ

In $\triangle ABC$ (#16), $AB = 11$ cm, $AC = 6$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 14 cm

B $\sqrt{223}$ cm

C $\sqrt{157}$ cm

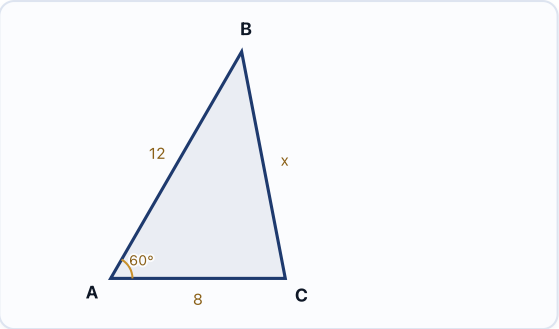
D $\sqrt{91}$ cm ✓

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 11^2 + 6^2 - 2(11)(6)\cos 60^\circ = 121 + 36 - 66 = 91$, so $BC = \sqrt{91}$ cm.

25 MCQ

In $\triangle ABC$ (#17), $AB = 12$ cm, $AC = 8$ cm, and $\angle BAC = 60^\circ$. Using the Cosine Rule, find the exact length of BC.



A 17 cm

B $\sqrt{208}$ cm

C $\sqrt{112}$ cm ✓

D $\sqrt{304}$ cm

WORKED SOLUTION & VERIFICATION

By the Cosine Rule, $BC^2 = 12^2 + 8^2 - 2(12)(8)\cos 60^\circ = 144 + 64 - 96 = 112$, so $BC = \sqrt{112}$ cm.